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A simplified pavement condition index regression model for pavement evaluation

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ABSTRACT

International Roughness Index (IRI) and Pavement Condition Index (PCI) are among other pavement condition indices used to assess pavement surface condition. The literature suggests that most of the pavement indices are related as a result of which several models have been developed to predict one index from the other. This study uses the Long-Term Pavement Performance (LTPP) database to develop a simplified regression model that links PCI with IRI. Measured pavement distresses from 1448 LTPP sections from the Specific Pavement Studies (SPS) and General Pavement Studies (GPS) representing 12744 data points were utilised for the PCI estimation. A total of 1208 sections with 10868 data points were used for model development while 240 sections with 1876 data points were used for the model validation. A sigmoid function is found to best express the relationship between PCI and IRI with a coefficient of determination (R^2) of 0.995. The bias in the predicted IRI values is significantly very low. The model validation using a different dataset also yielded highly accurate predictions ($R^2 = 0.992$). Finally, a pavement condition rating based on IRI is proposed. This system yields rating equivalent to the widely used PCI rating method which is based on the pavement condition.

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Pavement performance; long Term pavement performance; International roughness Index; pavement condition Index; regression analysis; pavement management

Introduction

The existing road network in any country is a key element to its growth. With time and exposure to traffic loading and environmental conditions, pavements continuously deteriorate. Thus, pavement maintenance and rehabilitation decisions represent a major challenge for highway agencies around the world. To overcome this challenge, many highway agencies have established procedures and practices for the purpose of preserving their pavement network by making the proper maintenance decision at the right time. Such practices are normally constrained by many factors such as lack of data, staffing, funding, and lack of expertise (Wotring 1998).

Reliable and precise assessment of the existing pavement network condition is a very important component of a successful pavement management system (PMS). This assessment has historically been achieved through an annual visual pavement condition inspection. For instance, pavement surface cracking is evaluated by a Surface Rating and Dominant Defects for each segment of the pavement network (ASTM 2007). However, the complete condition and performance of pavement are broader than just an assessment of the surface distresses. Other factors, such as ride quality, structural capacity and friction are also important components for a successful PMS. Currently, ride quality is considered as a basic element of pavement performance and customer satisfaction. New technologies are now available to measure other important pavement distresses at the network level. Imaging of the pavement surface through photographing or digitising and measuring the pavement profile using non-contact sensors are examples of such technologies (Mascio *et al.* 2007). Wang *et al.* (2015) used 3D laser imaging-based sensors

for pavement distress survey. This technology produces a longitudinal profile for roughness and a transverse profile for rutting. It also predicts the hydroplaning speed for safety analysis as well as various surface defects for distress evaluation.

The evaluation of pavement performance using pavement condition indicators is also a basic component of any PMS. Various indicators such as Pavement Condition Rating (PCR), Pavement Condition Index (PCI), Present Serviceability Rating (PSR), International Roughness Index (IRI), etc. have been commonly used to assign maintenance strategies for the existing pavements (Shah *et al.* 2013).

The PCI method is the most widely used index for pavement condition assessment throughout the United States and Canada. It is a comprehensive measure of the present pavement condition that is based on the observed surface distresses and sound statistical analysis for pavement sampling. It also indicates the pavement structural integrity and surface operational condition (Shahin and Kohn 1979, ASTM 2007). However, it is difficult in application, as it examines 19 different distresses with different severity levels (low, medium and high) and extent. Data for the PCI determination are collected through either visual inspection or image-based survey methods. The visual inspection takes a long time, and causes traffic interruption. It is also impractical for long roads and large road networks, and may be unsafe for workers doing the inspection. On the other hand, the image-based survey methods which use a vehicle to collect film, video, or digital images of the pavement system is faster and safer but requires advanced equipment and may be costly.

One of the major parameters affecting the ride quality and hence the user perspective about the road is the pavement

Table 1. Pavement Condition Criteria.

Country	Road Type	Acceptable Initial IRI (m/km)
Dubai (ISO 2016)	Freeways	0.90
	Arterials	1.00
	Collectors	1.20
Australia (Australroads 2007)	Freeways	1.60
	Highways and main roads	1.90
Belarus (MTiKRB 2012)	Highways and first class roads	1.50
	Second and third class roads	2.00
	Fourth and fifth class roads	2.50
Hungary (MUT 2008)	Main roads	1.20
	Primary and secondary roads	1.80
	Local roads	2.20
Norway (NPRA 2014)	Primary roads	2.00
	Other roads	2.50
Russia (MRRRF 2012)	Highways and first class roads	2.20
	Second and third class roads	2.20
	Fourth and fifth class roads	2.60
Slovakia (MDVaRR 2012)	Highways and expressways	1.90
	Primary and secondary roads	1.90
	Third class and local roads	3.30

roughness. The increase in pavement roughness increases fuel consumption and greenhouse gas emissions and decreases vehicle efficiency, and it may result in traffic safety issues that could lead to the loss of millions of dollars every year. Pavement roughness can be quantified through the International Roughness Index (IRI). IRI was developed by the World Bank in the 1980s (ARA, ERES Consultants, 2004). It is defined as ‘the accumulated suspension vertical motion divided by the distance traveled as obtained from a mathematical model of a simulated quarter-car traversing a measured profile at 80 km/h’ (ARA, ERES Consultants, 2004).

Robbins (2016) conducted a study quantifying the IRI significance regarding the increase in vehicle operating costs, fuel consumption, tire wear costs, vehicle maintenance and repair cost, and oil consumption. Moreover, many highway agencies around the world consider the initial IRI value as a quality assurance criterion while terminal IRI as an indicator of pavement maintenance actions or reconstruction needs (Perera and Kohn 2002, Robbins 2016). The Mechanistic-Empirical Pavement Design (MEPDG) considers the terminal IRI as one of the main pavement structural design criteria (ARA, ERES Consultants, 2004). Moreover, many agencies around the world present technical specifications to assess road condition based on IRI values. Table 1 presents the allowable initial IRI (IRI just after construction) values for different road functional classifications in different countries. Table 2 shows the pavement condition categorisation based on IRI values to control the quality of the road network from different studies and countries.

Mactutis *et al.* (2000) investigated the relationship between IRI, rutting and cracking based on 317 observations from roads located near Reno, Nevada which resulted in a coefficient

of determination (R^2) of 0.71. Lin *et al.* (2003) analyzed the relationships between IRI and pavement distresses using a back propagation artificial neural network (ANN) methodology. Their ANN was constructed using 14 nodes at the input layer, 6 nodes at the hidden layer, and one output. Yousefzadeh *et al.* (2010) discussed the capability of using ANN for road profile estimation. An ANN was developed by Vidya *et al.* (2013) to estimate IRI from PCI based on data obtained from construction work zones. Their model was constructed with three inputs, one hidden layer with four neurons and one output. The predicted IRI values from the model were compared with the actual measured IRI values for evaluating roughness using low-cost instrumentation within the construction work zones. The developed ANN model resulted in R^2 of 0.86 which indicates satisfactory predictions.

The MEPDG uses an IRI model for flexible pavements that was developed based on the Long-Term Pavement Performance (LTPP) database. The model predicts IRI as a function of initial IRI (IRI_o) which measured just after construction, site factor (SF), area of fatigue cracking (FC_{Total}), length of transverse cracking (TC) and average rut depth (RD) (ARA, ERES Consultants, 2004). This model was based on 1926 LTPP observations with R^2 of 0.56. In a recent work, Abd El-Aziz *et al.* (2017, 2018) presented regression and ANNs models predicting (IRI) as a function of distresses based on comprehensive LTPP database of 506 sections with 2439 observations. The models were proposed for IRI prediction as a function of age, IRI_o , all severities transverse cracks length ($(T.C)_{all}$), all severities alligator fatigue cracks area ($(F.C)_{all}$) and standard deviation of the rut depth (SDRUT). The regression analysis yielded fair accuracy with R^2 of 0.57 while the ANNs model yielded better accuracy with R^2 of 0.75.

The presented studies showed that IRI and PCI are both function of the pavement distresses. Thus, there should be a relationship between IRI and PCI. Few researchers explored the IRI-PCI relationship with acceptable statistical validity. Park *et al.* (2007) established a power relationship between PCI and IRI using data from nine states and provinces in Northern America. The IRI-PCI data was extracted from the DataPave programme for highways in the regions of Delaware, Maryland, New Jersey, New York, Vermont, Virginia, Ontario, Quebec, and Prince Edward Island. The database includes 20 pavement sections with 63 data points. They proposed a power model correlating the IRI (in units of m/km) with the PCI as presented in Equation (1).

$$\log(PCI) = 2 - 0.436 \log\left(\frac{IRI}{0.727}\right) \quad (1)$$

The accuracy of the model was determined to be fair ($R^2 = 0.59$). Dewan and Smith (2002) presented another model

Table 2. Pavement Condition Categorisation based on IRI.

Pavement Quality	Sayers <i>et al.</i> (1986)	FHWA (2003)		Acceptable IRI (m/km)		
		Interstates	Other	Cantisani and Loprencipe (2010)	INVIAS Specifications (2007)	Goenaga <i>et al.</i> (2017)
Very good	< 2.00	< 1.00	< 1.00	< 1.42	2.00–3.50	< 2.80
Good	2.00–3.50	1.00–1.50	1.00–1.50	1.42–2.84	3.50–4.50	2.80–3.50
Fair	3.50–6.00	1.50–1.90	1.50–2.68	2.84–4.06	4.50–6.50	3.50–4.30
Poor	> 8.00	> 2.70	> 3.47	> 4.06	> 6.50	> 4.30

Equation (2) for IRI as a function of PCI for the Bay Area cities and counties in California with the intent of using the model in estimating user costs/benefits for their PMS.

$$\text{IRI} = 0.0171(153 - \text{PCI}) \quad (2)$$

The R^2 was 0.53 with a coefficient of variation of 28% which represents a poor correlation. In another work, Arhin *et al.* (2015) proposed models predicting PCI using IRI for a dense urban area by functional classification and pavement type in the District of Columbia. The models were determined to be statistically significant with R^2 values between 0.56 and 0.82. The relation between PCI and IRI (in/mile) for flexible pavements is presented in Equation (3).

$$\text{PCI} = -0.224 (\text{IRI}) + 120.02 \quad (3)$$

Regardless of the great effort that has been exerted, the literature models correlating PCI with IRI were based only on a limited data that did not cover a wide range of IRI and PCI values representing the wide and variable range of pavement conditions. In addition, the accuracy of these models was mostly poor to fair.

Objectives

The pavement condition survey required for PCI determination is time-consuming and costly as compared to IRI. In addition, since IRI is affected by various pavement distresses as reported in the literature (Mactutis *et al.* (2000), Lin *et al.* (2003), Abd El-Aziz *et al.* (2017, 2018)), the hypothesis of this research is that there should be a strong relationship between IRI and PCI. Thus, the main objective of this research is to develop an accurate and reliable IRI-PCI model based on a large and reliable database representing the wide range of pavement structures exposed to different climatic conditions and traffic levels. Another objective of this paper is to propose a pavement rating system for pavement management based on IRI values corresponding to specific pavement surface conditions. This rating system should yield equivalent rating as the existing well-known PCI rating system.

Database collection and processing

The LTPP database is one of the most comprehensive and reliable sources of pavement performance data. It was developed as a part of the Strategic Highway Research Program (SHRP) in 1987 (DataPave 2001). The LTPP database covers different states in America and Canada with different pavement structures and properties, ages, climatic conditions, and traffic levels. LTPP data includes general inventory and information of test sections, material properties, maintenance and rehabilitation (M&R), climate, traffic, deflection, longitudinal profile (IRI) and pavement distresses. For this research, the LTPP standard data release number 30 (released in October 2016) is used (LTPP InfoPave). In general, there are two main experiments in the LTPP, Specific Pavement Studies (SPS) and General Pavement Studies (GPS). SPS are studies of specially constructed, maintained, or rehabilitated pavement sections incorporating a controlled set of experimental design and construction features. GPS consists of a series of studies on nearly 800 in-service

pavement test sections throughout the United States and Canada. The fundamental difference between these two classifications is that at the start of the LTPP program, the GPS test sections were existing pavements while the SPS projects are sites where multiple test sections of different experimental treatment factors are constructed. The collected data was extracted from four modules; namely ADM, MON, SPS and INV. These modules and the corresponding data tables are illustrated in Table 3.

The final database used in this study consists of 1448 test sections with 12,744 observations, including 1208 LTPP-SPS sections with 10,868 (85%) data points for model development and 240 LTPP-SPS and GPS sections with 1876 (15%) data points for model validation. The geographic distribution of the SPS and GPS test sections used in the analysis is shown in Figure 1.

In the LTPP database, the roughness data is reported in meter per kilometer (m/km). These measurements were recorded using vehicles equipped with sensors to detect the longitudinal profile variation of the pavement. The roughness data was measured in the left and right wheel paths. The extracted data contains State Code, SHRP_ID, date of measurement, and the average of the two-wheel path readings which was used as the IRI value for a particular measurement date as shown in Table 4.

The distress database in the LTPP programme consists of individual distress data of asphalt concrete pavements (AC) as well as rigid pavements. In LTPP, 13 common distresses are considered for flexible pavements. Each entry dataset includes inventory and general information (State Code, Section/SHRP_ID), in addition to distress information and survey dates. A test section is generally 3.7 m (12 ft) wide $\times$ 152.4 m (500 ft) long with an area of 563.88 m² (6000 ft²). The distress data includes extent, type, and severity as shown in Table 4.

The investigated pavement structures are flexible pavements with asphalt concrete layer(s) having thicknesses between 1.5 in (3.8 cm) to 15 in (38 cm) over the unbound granular base with a thickness between 3.8 in (9.65 cm) to 31.4 in (80 cm). Some investigated sections contain bound treated base with a thickness between 3.1 in (7.87 cm) to 16.4 in (42 cm).

In order to overcome the variation between measured pavement distresses survey date and visit date for pavement roughness measurement, this research followed a similar procedure used in MEPDG. Firstly, a regression model correlating the measured IRI and corresponding pavement age were developed for each section considering the traffic opening date as the

Table 3. LTPP Database Used Modules and Tables.

Data	LTPP Module	LTPP Table
General Information	ADM	EXPERIMENT_SECTION SECTION_LAYER_STRUCTURE
IRI	MON	MON_HSS_PROFILE_SECTION MON_HSS_RUN_NO MON_HSS_VISIT_NO MON_DISC_AC_REV
Pavement Distresses		MON_T_PROFILE_INDEX_SECTION
Rutting		SPS_ID
SPS General Information	SPS	
Dates	INV	INV_AGE

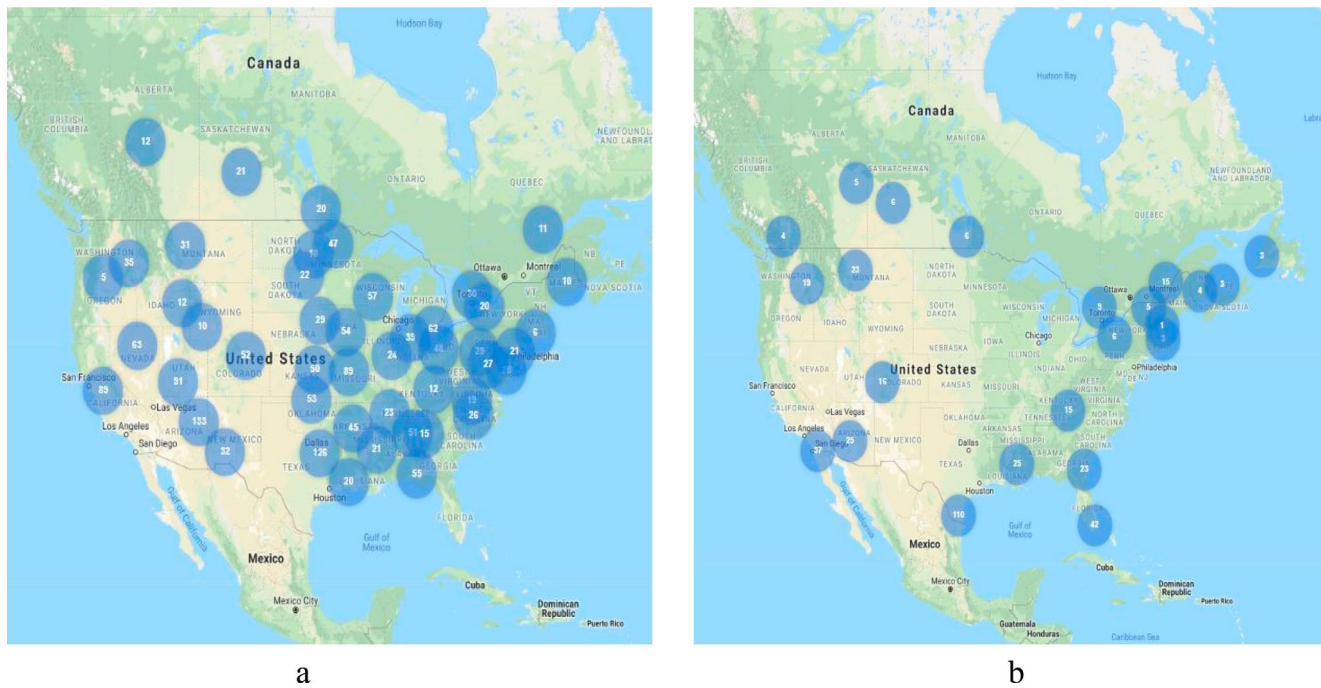


Figure 1. Locations of the LTPP Sections used in this Study, (www.infopave.com). a. SPS Sections Locations. b. GPS Sections Locations.

starting age (ARA, ERES Consultants, 2004). For sections with effective maintenance activities (activities which significantly affected the value of IRI), only the portion of the measured IRI data from the first IRI measurement until the first drop of the IRI value at the time of maintenance was considered (Abd El-Aziz *et al.* 2017). The developed regression models were then used to backcast the IRI value as a function of pavement age through linear or exponential fit as recommended in Abd El-Aziz *et al.* (2017, 2018).

Figure 2 exemplifies linear and exponential fit for two different LTPP sections along with the regression function and the coefficient of determination (R^2). These models were also used to estimate the IRI values at the dates of distress measurements as suggested by other researchers (Abd El-Aziz *et al.* 2017, 2018).

The IRI data versus age for each section was drawn to examine and remove any erroneous data from the database such as

sections with IRI data following illogic behaviour, i.e. decrease in IRI with pavement age as shown in Figure 3.

Pavement condition index calculation

PCI is a numerical index between 0 and 100 used to express the general condition of the surface of a pavement section, with 100 representing the best possible condition and 0 representing the worst possible condition (ASTM D 6433-07, 2007). After getting the distress data and the corresponding severities from the LTPP database, the PCI was calculated following the ASTM D 6433-07 procedure (ASTM D 6433-07, 2007). In this procedure, the PCI is calculated from Equation (4) as a function of the corrected deducts values (CDV).

$$PCI = 100 - \max CDV \quad (4)$$

Table 4. Example of the IRI and Distress Data Extracted from the LTPP Database.

State Code	SHRP_ID	VISIT_DATE	Average IRI	SURVEY_DATE	GATOR_CRACK_A_L	GATOR_CRACK_A_M	GATOR_CRACK_A_H	TRANS_CRACK_L_L	TRANS_CRACK_L_M	TRANS_CRACK_L_H
1	0101	30/10/1995	0.657	25/08/1994	0	0	0	0	0	0
1	0101	10/01/1996	0.667	08/02/1995	0	0	0	0	0	0
1	0101	08/04/1996	0.682	26/07/1995	0	0	0	0	0	0
1	0101	12/08/1996	0.697	16/04/1996	0.6	0	0	0	0	0
1	0101	11/10/1996	0.678	10/10/1996	1.2	0	0	0	0	0
1	0101	03/07/1997	0.687	30/10/1997	0	0	0	0	0	0
1	0101	15/10/1997	0.702	25/04/1998	0	0	0	0	0	0
1	0101	27/01/1998	0.679	17/11/1998	3	0	0	0.3	0	0
1	0101	23/04/1998	0.694	18/05/2000	38.6	0	0	0.6	0	0
1	0101	05/08/1998	0.810	23/08/2001	21.8	0	0	0.8	0	0
1	0101	07/12/1998	0.768	08/02/2002	31.1	0	0	3.5	0	0
1	0101	28/09/1999	0.714	09/04/2003	64.9	0	0	13.6	1	0
1	0101	14/03/2001	0.743	23/02/2004	68	0	0	14.7	1	0
1	0101	10/03/2002	0.765	28/04/2005	70.4	0	0	13.6	1	0

A_L: Area of alligator (fatigue) cracking of low severity, m^2 , A_M: Area of alligator (fatigue) cracking of moderate severity, m^2 , A_H: Area of alligator (fatigue) cracking of high severity, m^2 , L_L: Length of low severity transverse cracking, m, L_M: Length of moderate severity transverse cracking, m, L_H: Length of high severity transverse cracking, m.

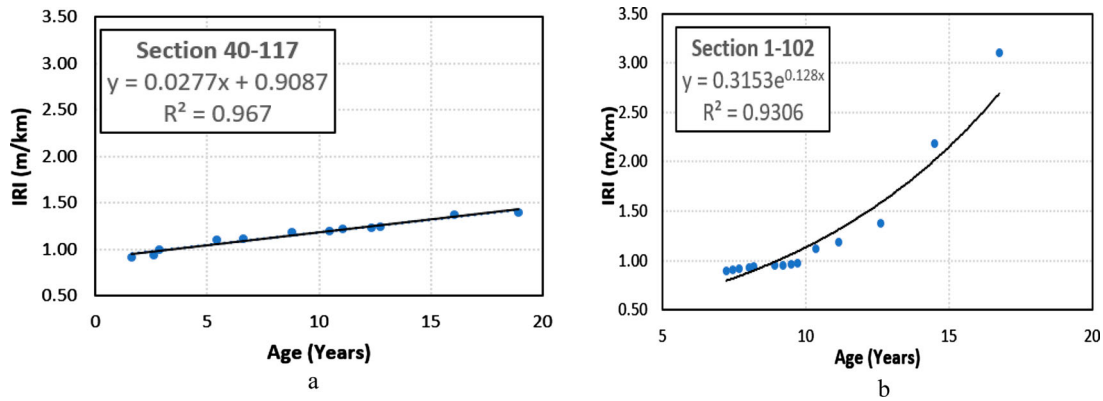


Figure 2. Linear and Exponential Fit for Two Different LTPP Sections. a. Linear Fit. b. Exponential Fit.

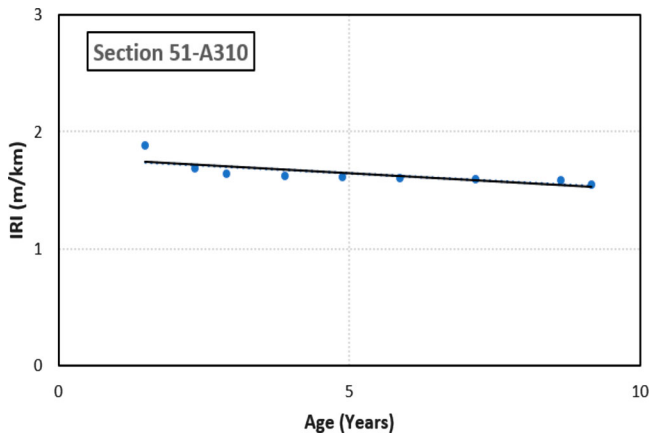


Figure 3. Example of Erroneous IRI Data.

For each distress, the deduct value (DV) is calculated based on the severity and density of the distress. Then, the ASTM procedure is followed to correct the deduct values. Doing this manually would take a very long time especially for a large database as the one under consideration. Using the available software such as the Micro Paver still also be

Table 5. Descriptive Statistics of IRI, PCI and Distress Data used for Model Development.

Variable	N	Mean	Standard Deviation	Standard Error	Min.	Max.
IRI (m/km)	10868	2.58	1.69	0.016	0.61	5.19
PCI	10868	60.85	31.58	0.303	6.10	100
Fatigue Cracking (m ²)	—	9.70	40.81	0.391	0	816.6
Edge Cracking (m)	—	0.33	5.75	0.059	0	152.5
Block Cracking (m ²)	—	13.53	73.24	0.658	0	603.9
Longitudinal Cracking (m)	—	8.12	25.57	0.230	0	307.1
Transverse Cracking (m)	—	27.14	41.63	0.370	0	529.4
Patching (m ²)	—	4.97	31.41	0.280	0	530.4
Potholes (m ²)	—	0.01	0.09	0.001	0	7.15
Shoving (m ²)	—	0.07	2.68	0.030	0	152.5
Bleeding (m ²)	—	14.89	61.44	0.600	0	610
Polished	—	2.21	25.96	0.250	0	557.8
Aggregate (m ²)	—	—	—	—	—	—
Rutting (mm)	—	4.56	3.13	0.030	0	29
Raveling (m ²)	—	34.35	117.49	1.160	0	640.5

The data shown in per section.

time-consuming as the distress data for each section has to be entered manually. Thus, an excel sheet was developed using the procedure suggested by Kan Wu *et al.* (2015) to automate the calculation of the PCI values for the entire database. A comparison of the descriptive statistics for both IRI, PCI, and LTPP distresses used for the model development is presented in Table 5.

IRI-PCI model development

After calculating the PCI for all sections, the computed PCI values were plotted against the corresponding measured IRI data for all sections as shown in Figure 4. The IRI-PCI shows an S-shaped curve resembling the sigmoid mathematical form given in Equation (5). This sigmoid function is being used in many applications in the pavement and geotechnical fields. The most widely used application of the sigmoid function in the pavement field is the Hot Mix Asphalt (HMA) dynamic modulus (E^*) master curve (El-Ashwah *et al.* 2019) and modelling resilient modulus of soils and unbound materials taking into account the effect of moisture variation (AASHTO, 2008, El-Ashwah *et al.* 2019). Other applications of the sigmoid function are the modelling of pavement performance (Beckley *et al.* 2016) and the soil water characteristics curve (Johari *et al.* 2011).

$$IRI = \frac{K_1}{K_2 + \exp(K_3 * PCI)} \quad (5)$$

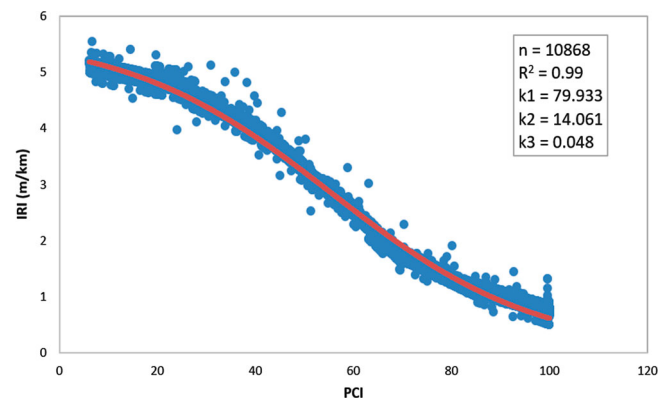


Figure 4. Relationship Between IRI and PCI Values Equation (5).

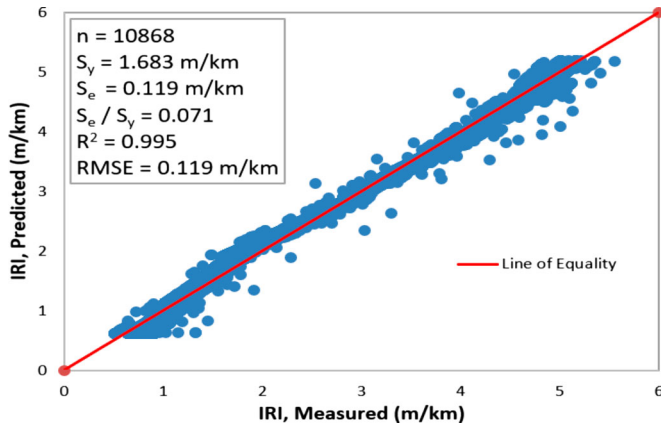


Figure 5. Relationship Between Measured and Predicted IRI Values of the Proposed Model.

where, IRI = Predicted International Roughness Index, m/km. PCI = Pavement Condition Index. K_1 , K_2 , K_3 = regression parameters describing the shape of the sigmoid function.

This model is determined by minimising the sum of square errors of predicted IRI versus observed IRI by changing the regression parameters of the sigmoidal function using non-linear optimisation through the Excel Solver add-in. In order to assess the accuracy of this relationship, the goodness of fit statistics was calculated as given in Eqs. (6 through 10).

$$S_y = \sqrt{\frac{1}{n} \sum_{i=1}^n (IRI_{(m)} - IRI_{(average)})^2} \quad (6)$$

$$E = \sum_{i=1}^n (IRI_{(m)} - IRI_{(p)}) \quad (7)$$

$$S_e = \sqrt{\frac{\sum_{i=1}^n E^2}{n - p}} \quad (8)$$

$$R^2 = \left[1 - \left[\frac{n - p}{n - 1} \right] * \left[\frac{S_e}{S_y} \right]^2 \right] \quad (9)$$

$$RMSE = \sqrt{\sum_{i=1}^n \left[\frac{(IRI_{(m)} - IRI_{(p)})^2}{n} \right]} \quad (10)$$

where, S_y = standard deviation of the measured values about the mean value. E = sum of errors between predicted and measured IRI. $IRI_{(m)}$ = Measured International Roughness Index. $IRI_{(average)}$ = Average International Roughness Index. $IRI_{(p)}$ = Predicted International Roughness Index. S_e = standard deviation of the error. n = number of total data points; p = number of regression coefficients. R^2 = coefficient of determination. RMSE = root mean square error.

S_e/S_y and R^2 represent the degree of scattering with respect to the line of equality. S_e/S_y values are good if less than 0.5; and marginal if greater than 0.75 (Beckley *et al.* 2016). R^2 ranges from 0 to 1. A higher coefficient indicates better goodness of fit. RMSE is a frequently used measure of the differences

between values predicted by a model and the values actually observed. Lower values of RMSE indicate a better fit.

The relationship between predicted and measured IRI using the proposed sigmoid function along with the goodness of fit statistics is depicted in Figure 5.

The sigmoidal function with the final regression constants is presented in Equation (11).

$$IRI_{(p)} = \frac{79.933}{14.061 + \exp(0.048 * PCI)} \quad (11)$$

As shown in Figure 5, the model produces highly accurate predictions as indicated by the very high R^2 of (0.995), very low S_e/S_y of 0.071 and low RMSE of 0.119 m/km. Four different LTPP sections are selected randomly to examine the model prediction accuracy at different ages for the same section. Figure 6 exemplifies the proposed sigmoid model (shown as a solid line) and the measured IRI-PCI values at different times (shown as dots) for the same section along with the R^2 . This figure confirms the prediction capability of the model.

Residuals (errors) and their normal scores confirm the normality and the homoscedasticity of the distribution. Moreover, the mean error value is very small with a small standard deviation. In addition, the bias in the predicted IRI values is significantly very low as shown in Figures 5 and 7.

Model validation

A total of 240 different LTPP – GPS and SPS sites with 1876 data points were selected randomly from different climatic areas to validate the developed model. Table 6 summarises the descriptive statistics of the 1876 data points used for model validation. Comparison of the data used for the model development shown previously in Table 5 and the data used for the model validation (shown in Table 6) reveals that the validation data has almost the same ranges of data used for the model development. Figure 8 shows the final results of measured and predicted IRI values using the developed model. It is evident from the figure that the developed model provides an acceptable IRI prediction. The figure shows that the model produces highly accurate predictions with R^2 of 0.992 and a minimal bias indicating a highly accurate and powerful model.

Pavement condition rating using IRI and PCI

Using Equation (11), the IRI ranges corresponding to PCI ranges that express different pavement condition ratings can be determined. Figure 9 displays the overall range of the IRI values and the predicted pavement condition ratings. Alternatively, IRI may be assessed as a predictor variable of functional pavement condition.

The values presented in Figure 9, show the relationship between IRI and corresponding PCI for different pavement conditions. Figure 10, illustrates a comparison between the IRI ranges for the different pavement conditions resulted from the proposed model based on the corresponding PCI values with the IRI ranges found in the literature. The values presented in Figure 10, show that the IRI rating values from

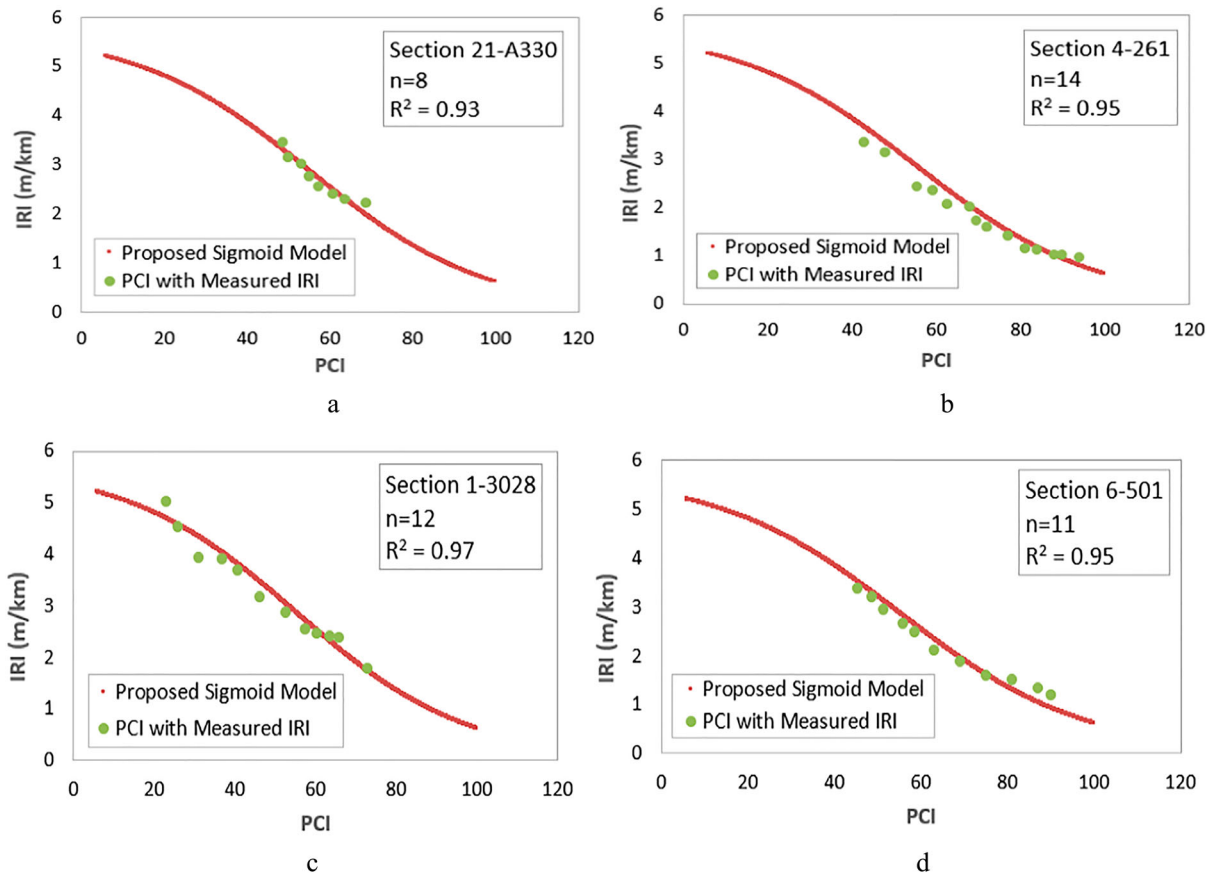


Figure 6. IRI Versus PCI at Different Ages for Different LTPP Sections. a. LTPP Section 21-A330. b. LTPP Section 4-261. c. LTPP Section 1-3028. d. LTPP Section 6-501.

the proposed model reasonably agree with IRI values related to some literature studies. For example, Sayers *et al.* (1986), FHWA (2003), Cantisani and Loprencipe (2010), and Goenaga *et al.* (2017) recommended acceptable IRI for very good pavement condition to be less than 2.0, 1.0, 1.42, and 2.80 m/km, respectively while the developed model recommends IRI value less than 1.90 m/km for the same condition. The importance of the proposed rating system is that, based on IRI values, the pavement condition can be determined. Even though this pavement condition is determined based on the surface roughness, it still reflects the surface distresses as explained in the next section.

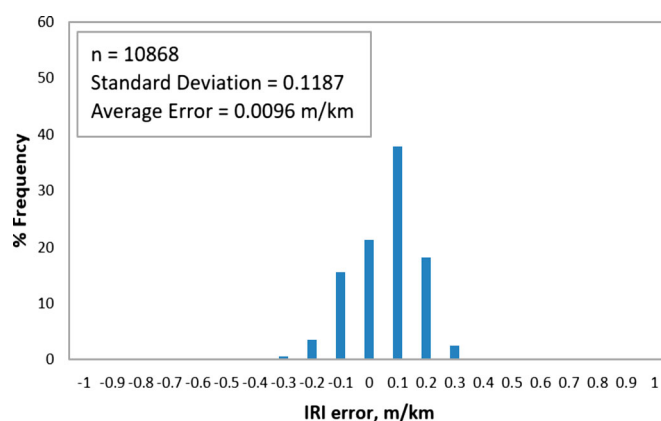


Figure 7. Normal Error Distribution.

Distress correlation analysis

The selection of the right maintenance strategy at the right time is of utmost importance for the effective management of highway pavements. The appropriate maintenance strategy is influenced by type, severity, and extent of the pavement surface distresses and the structural and roughness condition of the pavement. Most researchers in the field of PMS impose

Table 6. Descriptive Statistics of IRI, PCI and Distress Data used for Model Validation.

Variable	N	Mean	Standard Deviation	Standard Error	Min.	Max.
IRI (m/km)	1876	2.77	1.67	0.038	0.64	5.21
PCI	1876	57.34	31.32	0.723	6.17	100
Fatigue Cracking (m ²)	–	14.13	33.48	0.631	0	371.7
Edge Cracking (m)	–	1.69	11.73	0.323	0	152.5
Block Cracking (m ²)	–	8.75	57.89	1.595	0	564.3
Longitudinal Cracking (m)	–	8.68	24.33	0.670	0	305
Transverse Cracking (m)	–	21.36	36.99	1.019	0	287.4
Patching (m ²)	–	2.14	18.06	0.390	0	286.4
Potholes (m ²)	–	0.01	0.06	0.002	0	2.99
Shoving (m ²)	–	0.04	1.42	0.039	0	51.4
Bleeding (m ²)	–	14.61	59.46	1.801	0	372.5
Polished	–	2.07	28.50	0.750	0	503.3
Aggregate (m ²)	–	–	–	–	–	–
Rutting (mm)	–	4.11	2.54	0.070	0	20
Raveling (m ²)	–	24.18	97.49	1.950	0	579.5

The data shown in per section.

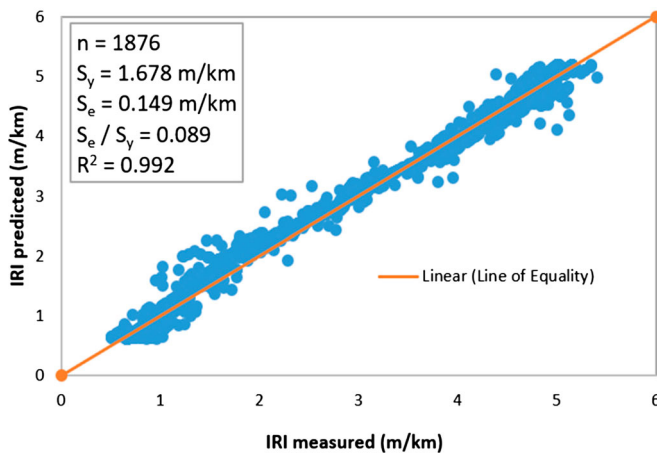


Figure 8. Relationship of the Measured and Predicted IRI Values for Model Validation.

maintenance strategies including: no maintenance, minor maintenance, major maintenance, and full reconstruction without knowing the distresses found in pavement sections. This may lead to irrational decisions such as over distinctive cost or under distinctive cost (Morcous and Lounis (2004), Chikezie *et al.* (2013), and Gao *et al.* (2012)). In this study, based on the comprehensive LTPP data collected, the most probable distresses found for each pavement condition (excellent, very good, good, fair, poor, very poor, and failed) are identified and correlated with the corresponding IRI values.

In order to determine the most probable distress related to different IRI ranges, the LTPP-SPS data are filtered based on the IRI classification ranges from failed to excellent. The Statistical Package for Social Science (SPSS) software is used to conduct the required statistical analysis. A total of 10,868 LTPP distress points related to 1,208 sections are used in this analysis. Then,

PCI	Pavement Quality	IRI (m/km)
100	Excellent	0.60
85		1.10
70	Very Good	1.90
55	Good	2.85
40	Fair	3.85
25	Poor	4.60
10	Very Poor	5.10
0	Failed	5.30

Figure 9. Pavement Quality for given IRI or PCI Values.

IRI (Proposed Model)	Pavement Condition	IRI Sayers et al. (1986)	IRI Cantisani and Loprencipe (2010)	IRI FHWA (2003)	IRI Goenaga et al. (2017)
<1.90	Very Good	< 2.00	< 1.42	< 1.00	< 2.80
1.90-2.85	Good	2.00-3.50	1.42-2.84	1.00-1.50	2.80-3.50
2.85-3.85	Fair	3.50-6.00	2.84-4.06	1.50-2.68	3.50-4.30
> 3.85	Poor	> 8.00	>4.06	>3.47	>4.30

Figure 10. Comparison of Pavement Condition Rating based on IRI Values Equivalent to PCI using the Proposed Model and Literature Models.

Pearson correlation analysis is conducted in order to find the most correlated distresses with different IRI classification ranges.

The summary of the correlation analysis obtained is presented in Table 7. The results show the most correlated distresses for different IRI classification range based on Pearson correlation values. A significance level of 0.01 at a minimum

Table 7. Summary of Correlation Analysis.

Pavement Condition	IRI Ranges	Correlated Distresses	Pearson Correlation
Very Good	1.10–1.90	Alligator Cracking L	0.822
		Bleeding	0.407
Good	1.90–2.85	Alligator Cracking L	0.587
		Block Cracking L	0.552
		Longitudinal Cracking L	0.448
Fair	2.85–3.85	Transverse Cracking H	0.693
		Potholes L	0.587
		Alligator Cracking M	0.450
		Patch M	0.437
		Edge Cracking M	0.432
		Polished aggregate	0.417
Poor	3.85–4.60	Transverse Cracking H	0.671
		Rutting	0.596
		Potholes L	0.578
		Alligator Cracking M	0.479
		Transverse Cracking L	0.461
		Alligator Cracking H	0.451
		Raveling	0.444
		Polished aggregate	0.423
		Longitudinal Cracking L	0.418
Very Poor	4.60–5.10	Transverse Cracking H	0.680
		Rutting	0.558
		Potholes H	0.540
		Alligator Cracking M	0.472
		Patch H	0.424
		Transverse Cracking L	0.421
		Longitudinal Cracking M	0.417
		Block Cracking H	0.400
		Transverse Cracking M	0.402
		Polished aggregate	0.406
Failed	5.10–5.30	Longitudinal Cracking H	0.542
		Potholes H	0.552
		Alligator Cracking L	0.499
		Rutting	0.489
		Block Cracking H	0.469
		Transverse Cracking H	0.450
		Alligator Cracking H	0.436
		Alligator Cracking M	0.420
		Edge Cracking H	0.364
		Polished aggregate	0.361
		Patch H	0.364

L = Low Severity, M = Medium Severity, H = High Severity.

value of Pearson correlation of 0.360 is used as the criterion for the probability of the distress to exist. The importance of this analysis is that it helps the decision makers to suggest the best maintenance strategies based on the most probable distresses expected to lead to this specific pavement condition using the proposed IRI pavement rating system.

Summary and conclusions

This paper focused on the development of a relationship between IRI and PCI using the LTPP database for flexible pavements. A total of 1208 sections with 10868 data points were used for the model development and 240 sections with 1876 data points were used for the model validation. The following conclusions can be drawn from the present study:

- A sigmoid relationship was found between IRI and PCI. The proposed sigmoidal model produces accurate predictions as indicated by R^2 of 0.995, and Se/Sy of 0.071. The validation process indicates that the developed model can accurately be used for IRI predictions based on PCI data.
- The model developed in this study could be used to predict the overall range of IRI values corresponding to pavement condition ratings for pavement management assessments.
- An IRI-based pavement condition rating system was proposed which yields conditions equivalent to the ones proposed by the well-known PCI-based rating system.
- The presented distresses correlation analysis based on the comprehensive LTPP database allows pavement management engineers to select the appropriate maintenance strategies related to a specific pavement condition as a function of IRI instead of PCI.

Disclosure statement

No potential conflict of interest was reported by the authors.

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